Fig. 5 | Spatial context and reference-based evaluation of velocity directions. a, Spatial fields for 20 methods on the 608-spot MPSTA ST_small subset, ordered by descending GDC from left to right and top to bottom. The first four tiles show all 1,292 spots, the target spots, cell-type colors and five schematic reference transitions. TopoVelo GDC is 0.634 (607 nonzero gene-velocity predictions and one zero vector); TopicVelo GDC is 0.484 and uses native-transition displacement T @ Ms - Ms. **b,** Paired spatial-direction cosine maps for the five highest-GDC methods in a: TopoVelo, UniTVelo, TopicVelo, DeepVelo(VAE) and TFvelo. Colors indicate agreement between saved full-context and restricted-context directions; black crosses indicate undefined pairs. **c,** The same spatial-direction consistency across all 20 methods, ordered by median cosine. Boxes show the median and interquartile range; whiskers show the 5th–95th percentiles. TopoVelo has 606 valid pairs and other methods have 608. These within-dataset comparisons measure saved-workflow sensitivity, not an isolated causal effect of tissue context. **d,** KPTracer fields for PhyloVelo (backward), UniTVelo and DeepVelo(VAE), using the same 754 cells, fixed UMAP, equal aspect and shared streamline parameters. PhyloVelo retains its native backward arrows; evaluation uses the negated vectors to align the forward direction. **e,** KPTracer scores sorted by displayed value. The reference denominator is 706 cells in six states linked by four relations. PhyloVelo's displayed value of +1.000 is an adapted diagnostic: repeating its global gene-wise vector across cells makes within-state coherence structurally one. It is excluded from comparable GDC ranking and does not establish superior performance. The other 19 scores are unchanged. TopicVelo uses a native-transition displacement proxy.